

Machine learning classification – Random Forest algorithm

The following chart shows report that predicted by Confusion matrix

	Precision	Recall	F1-score	Support
0	0.93	0.92	0.92	85
1	0.86	0.88	0.87	49
Accuracy			0.90	134
Macro-average	0.89	0.90	0.90	134
Weighted-average	0.90	0.90	0.90	134

#Model prediction - [[78 7]

[6 43]]

Over all test set = 134

Non-purchased = 85 (T-78, F-7)

Purchased = 49 (T-43, F-6)

Actual class	Predicted class		
		Not-purchased	purchased
	(0)Not-purchased	True (0)	False (0)
	Purchased (1)	False (1)	True (1)

What is confusion matrix?

Confusion matrix is an evaluation metric that is used to evaluate model. In our case we worked for classification problem so that we use confusion matrix for evaluation. And we use some metric to evaluate model.

➤ Accuracy

Accuracy value shows the overall model performance. In other word **how many correct predictions made by our model out of our total given test input**

How we calculate it?

Accuracy = correct prediction / overall test input

Our models' accuracy= 0.92 # it made 121 (78+43) correct prediction out of 134

$$\text{Accuracy} = \frac{T(0) + T(1)}{T(0) + T(1) + F(0) + F(1)} = \frac{78+43}{134} = 0.92$$

➤ Recall (overall performance of specific class)

This metric shows % of True non-purchased out of total given value in test set

$$\text{Recall(non-purchased)} = \frac{T(n-p)}{T(n-p) + F(n-p)} \text{ (0.92)}$$

Same for purchased **(0.88)**

➤ Precision (is often used to find % of truly positive out of model thinks it's positive)

$$\text{Precision(N-P)} = \frac{T(n-p)}{T(n-p) + F(p)} \text{ (0.93)}$$

Same for precision (Positive) **(0.86)**

➤ **F1 measures**

Overall performance of non-purchased and purchased

$F1(n-p) = 2 * recall * precision / (recall + precision)$ **(0.92)**

$F1(p) = 2 * recall * precision / (recall + precision)$ **(0.87)**

➤ **Macro-average (% of recall, precision, and f1 measures)**

Macro-average (recall) = $(recall(n-p) + recall(p)) / 2$ **(0.89)**

Macro-average (precision) = $(precision(n-p) + precision(p)) / 2$ **(0.90)**

Macro-average (f1 meas) = $(f1\ meas(n-p) + f1\ measures(p)) / 2$ **(0.90)**

➤ **Weighted-average (sum of product of proportion rate of each class)**

Weighted-average(precision) = $precision * Total(np) / total\ test\ set + precision * Total(purchased) / total\ test\ set$

$(0.93 * (85/134) + 0.86 * (49/134)) = 0.90$

Same for recall = 0.90

Same for f1 measures = 0.90

